Timing is everything: investigating choroid plexus circadian dysfunction in hydrocephalus

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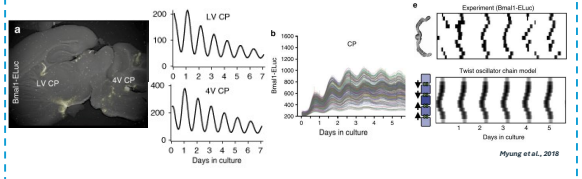
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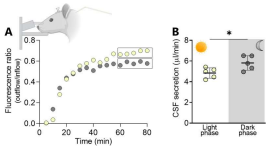
Introduction

Circadian rhythm disruption is a common phenomenon in neurological conditions and may accelerate disease progression. While the suprachiasmatic nucleus (SCN) is the master circadian clock, there has been increased interest in the circadian role of the choroid plexus (CP), a peripheral circadian oscillator that produces cerebrospinal fluid (CSF), given that CSF production and waste clearance are circadian. Previous studies have implicated CP dysfunction in fluid homeostasis impairment and hydrocephalus, which are characterized by impaired waste clearance and altered CSF dynamics, respectively. However, how changes in the SCN-CP phase relationships and CP oscillator coupling may lead to disrupted CP circadian organization, with potential implications for altered CSF dynamics, are largely unexplored.

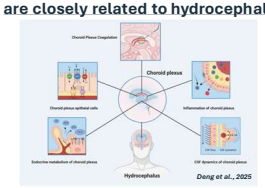
1) The CP is a circadian tissue with strong single-cell oscillations



2) CSF dynamics vary across light-dark phase



3) CP function and CSF dynamics are closely related to hydrocephalus

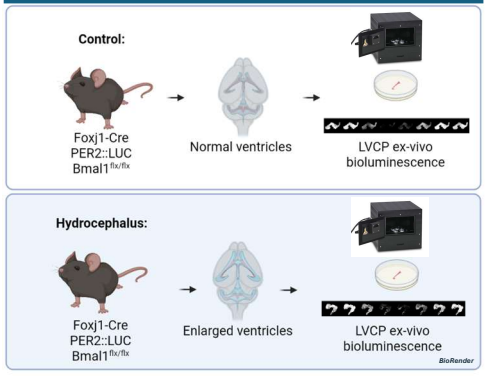


Research Question & Hypothesis

Do hydrocephalic mice exhibit altered circadian characteristics (period, amplitude) compared to non-hydrocephalic controls?

CPs from hydrocephalic mice will have a disruption of in period and amplitude regulation compared to nonhydrocephalic controls

Methods



Results

Whole-tissue choroid plexus rhythms from LumiCycle

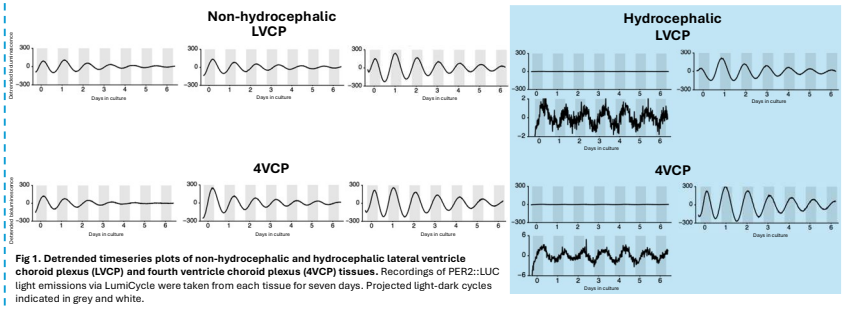


Fig 1. Detrended timeseries plots of non-hydrocephalic and hydrocephalic lateral ventricle choroid plexus (LVCP) and fourth ventricle choroid plexus (4VCP) tissues. Recordings of PER2::LUC light emissions via LumiCycle were taken from each tissue for seven days. Projected light-dark cycles indicated in grey and white.

Whole-tissue average period and amplitude

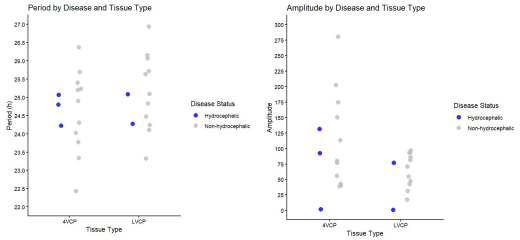


Fig 2. Overall period and amplitude based on tissue type and disease state. Recordings of PER2::LUC light emissions via LumiCycle were taken from each tissue for seven days, with each dot representing the average value of a tissue and colored by disease state.

Relationship between period & amplitude in LVCP from imaging

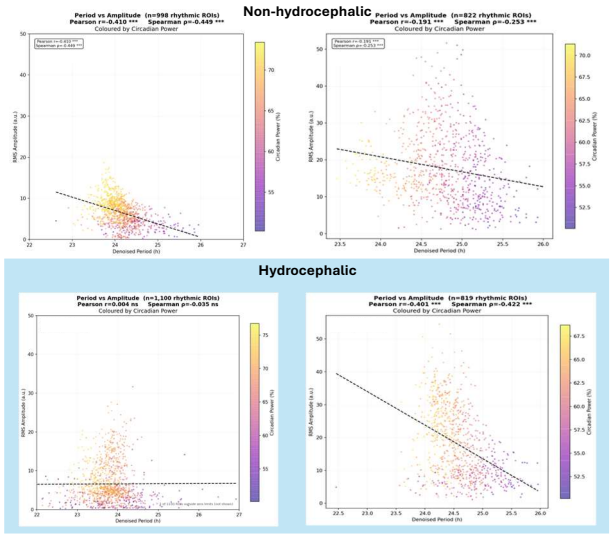


Fig 3. Single-ROI period vs. amplitude on non-hydrocephalic and hydrocephalic tissues. Recordings of Per2::LUC via bioluminescence imaging were captured over at least six days, with analysis excluding the first day. Each point represents one ROI from each tissue. Only rhythmic ROIs are shown (circadian power >50%, period between 22 and 26 h). Dashed line is ordinary least-squares linear fit.

Spatial organization of period and amplitude in LVCP

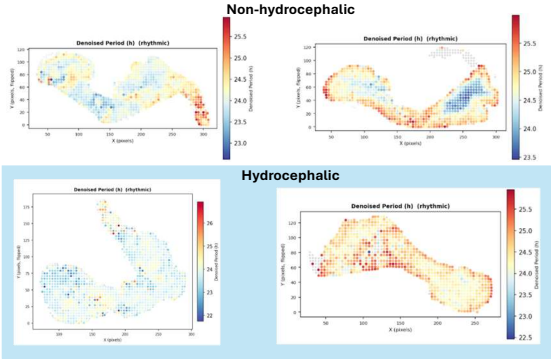


Fig 4. Spatial analysis of period in non-hydrocephalic and hydrocephalic LVCPs. Recordings of Per2::LUC via bioluminescence imaging were captured over at least six days, with analysis excluding the first day. Each dot represents one ROI at its physical position in the tissue. Outliers shown in grey.

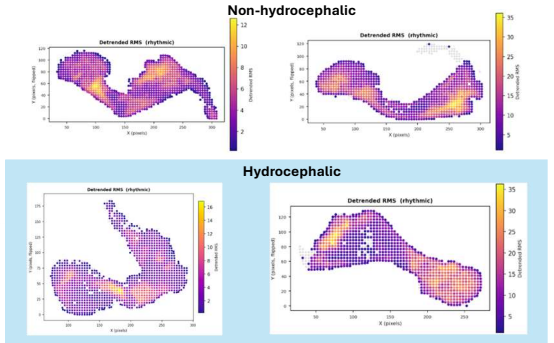


Fig 5. Spatial analysis of RMS amplitude in non-hydrocephalic and hydrocephalic LVCPs. Recordings of Per2::LUC via bioluminescence imaging were captured over at least six days, with analysis excluding the first day. Each dot represents one ROI at its physical position in the tissue. Outliers shown in grey.

Conclusion

In the hydrocephalus samples, while non-hydrocephalus control mice showed a negative correlation between period and amplitude, a subset of hydrocephalus mice had no correlation. Further testing may improve our understanding of the circadian role in choroid plexus dysfunction in hydrocephalus.

Future Directions

- Increase number of hydrocephalus samples and examine effect of tamoxifen
- Cluster distinct areas of the hydrocephalic CPs to determine if there are unique spatiotemporal circadian dynamics

Acknowledgements

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